

Multivariate Regression Analysis for Price Elasticity

1. Introduction

Pricing is one of the most critical levers in retail decision-making, directly influencing customer demand and revenue. Understanding how sensitive demand is to price changes—commonly referred to as price elasticity of demand—enables businesses to design effective pricing and promotional strategies.

This project applies multivariate regression analysis to estimate price elasticity using supermarket sales data. By controlling for customer behaviour, product categories, and store-level factors, the study aims to isolate the true impact of price on sales quantity. In addition, the project validates key regression assumptions and visualizes coefficient estimates with confidence intervals to ensure robustness and interpretability.

2. Objectives of the Study

The main objectives of this project are:

- To import and analyse sales and pricing data using R
- To build a multivariate linear regression model to estimate price elasticity of demand
- To evaluate price sensitivity across multiple product segments
- To validate regression assumptions, including multicollinearity and heteroscedasticity
- To visualize regression coefficients along with 95% confidence intervals
- To generate professional, reproducible outputs suitable for business and academic use

3. Project Structure and Reproducibility

To ensure clarity, reproducibility, and industry-standard workflow, the project follows a structured folder organization separating data, scripts, outputs, and reports.

	▲ Name	Size	Modified
	..		
☐	 .Rhistory	0 B	Jan 2, 2026, 4:26 PM
☐	 data		
☐	 outputs		
☐	 Price_Elasticity_Analysis.Rproj	218 B	Jan 2, 2026, 4:24 PM
☐	 report		
☐	 scripts		

Figure 1: Organized project folder structure used to ensure reproducibility and clean separation of data, scripts, outputs, and reports.

4. Dataset Description

4.1 Dataset Overview

The analysis uses a supermarket sales dataset containing 1000 transaction-level observations. The dataset represents realistic retail transactions and includes pricing, quantity sold, customer attributes, and product-level information.

The dataset is stored as `pricing_data.csv` and imported into R using tidyverse-compatible functions.

4.2 Dataset Preview

A preview of the dataset was performed to understand the structure and validate successful data import.

```
> head(data)
  sale_id branch      city customer_type gender product_name product_category unit_price
1       1     A   New York      Member   Male    Shampoo      Personal Care         5.50
2       2     B Los Angeles      Normal Female   Notebook      Stationery         2.75
3       3     A   New York      Member Female     Apple         Fruits          1.20
4       4     A   Chicago      Normal   Male    Detergent      Household         7.80
5       5     B Los Angeles      Member Female Orange Juice      Beverages         3.50
6       6     A   Chicago      Normal   Male    Shampoo      Stationery        11.24

  quantity  tax total_price reward_points
1         3 1.16      17.66             1
2        10 1.93      29.43             0
3        15 1.26      19.26             1
4         5 2.73      41.73             0
5         7 1.72      26.22             2
6         9 7.08     108.24             0
```

Figure 2: Preview of the supermarket sales dataset showing unit price, quantity sold, product category, and customer attributes.

4.3 Variable Description

Variable	Description	Analytical Role
quantity	Units sold per transaction	Dependent variable (Demand)
unit_price	Price per unit	Key independent variable
product_category	Product segment	Segmentation variable
reward_points	Loyalty incentive points	Control variable
customer_type	Member / Normal	Control variable
branch	Store branch	Control variable

Derived variables such as `total_price` were excluded from modelling to avoid multicollinearity.

5. Methodology

5.1 Model Selection

A multivariate log-log linear regression model was chosen for this study. The log-log specification allows regression coefficients to be interpreted directly as elasticities, which is a standard approach in demand analysis.

This approach enables clear economic interpretation while accounting for multiple demand-shifting factors.

5.2 Model Specification

The regression model is defined as:

$$\begin{aligned} \log(\text{Quantity}) &= \beta_0 + \beta_1 \log(\text{Unit Price}) + \beta_2(\text{Reward Points}) + \beta_3(\text{Customer Type}) + \beta_4(\text{Branch}) \\ &+ \beta_5(\text{Product Category}) + \epsilon \end{aligned}$$

Where:

- β_1 represents the price elasticity of demand
- Other coefficients capture the effects of loyalty, customer type, and segmentation

6. Regression Results

6.1 Multivariate Regression Output

The multivariate regression model was estimated using R's `lm()` function.

Call:

```
lm(formula = log_quantity ~ log_unit_price + reward_points +  
    customer_type + branch + product_category, data = data_clean)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.4459	-0.3579	0.1164	0.5237	1.2593

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2.030254	0.093784	21.648	<2e-16	***
log_unit_price	-0.314090	0.034916	-8.995	<2e-16	***
reward_points	0.065565	0.003474	18.871	<2e-16	***
customer_typeNormal	0.721312	0.060992	11.826	<2e-16	***
branchB	-0.025047	0.049087	-0.510	0.610	
product_categoryFruits	0.031313	0.072711	0.431	0.667	
product_categoryHousehold	-0.048987	0.073742	-0.664	0.507	
product_categoryPersonal Care	0.026480	0.072904	0.363	0.717	
product_categoryStationery	-0.089079	0.074001	-1.204	0.229	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.722 on 991 degrees of freedom

Multiple R-squared: 0.2718, Adjusted R-squared: 0.266

F-statistic: 46.25 on 8 and 991 DF, p-value: < 2.2e-16

Figure 3: Summary of the multivariate log-log regression model estimating price elasticity.

6.2 Price Elasticity Interpretation

The estimated coefficient for `log(unit_price)` is:

- Elasticity Estimate: -0.314
- Statistical Significance: p-value < 0.001

Interpretation:

A 1% increase in unit price leads to an approximate 0.31% decrease in quantity sold, holding other variables constant. This indicates price-inelastic demand, which is typical for essential supermarket products.

6.3 Model Fit

- **R-squared: 0.272**
- **Adjusted R-squared: 0.266**

The model explains approximately 27% of the variation in demand, which is reasonable and expected for cross-sectional retail transaction data involving heterogeneous consumer behaviour.

7. Model Validation

7.1 Multicollinearity Check

Multicollinearity was assessed using the Variance Inflation Factor (VIF).

```
> vif(elasticity_model)
```

	GVIF	Df	GVIF ^{1/(2*Df)}
log_unit_price	1.224321	1	1.106491
reward_points	2.022395	1	1.422109
customer_type	1.782075	1	1.334944
branch	1.015542	1	1.007741
product_category	1.022498	4	1.002785

Figure 4: VIF values indicating no multicollinearity issues among model predictors.

All VIF values were close to 1, confirming stable and reliable coefficient estimates.

7.2 Residual Diagnostics

Residual plots were examined to assess heteroscedasticity and overall model behaviour.

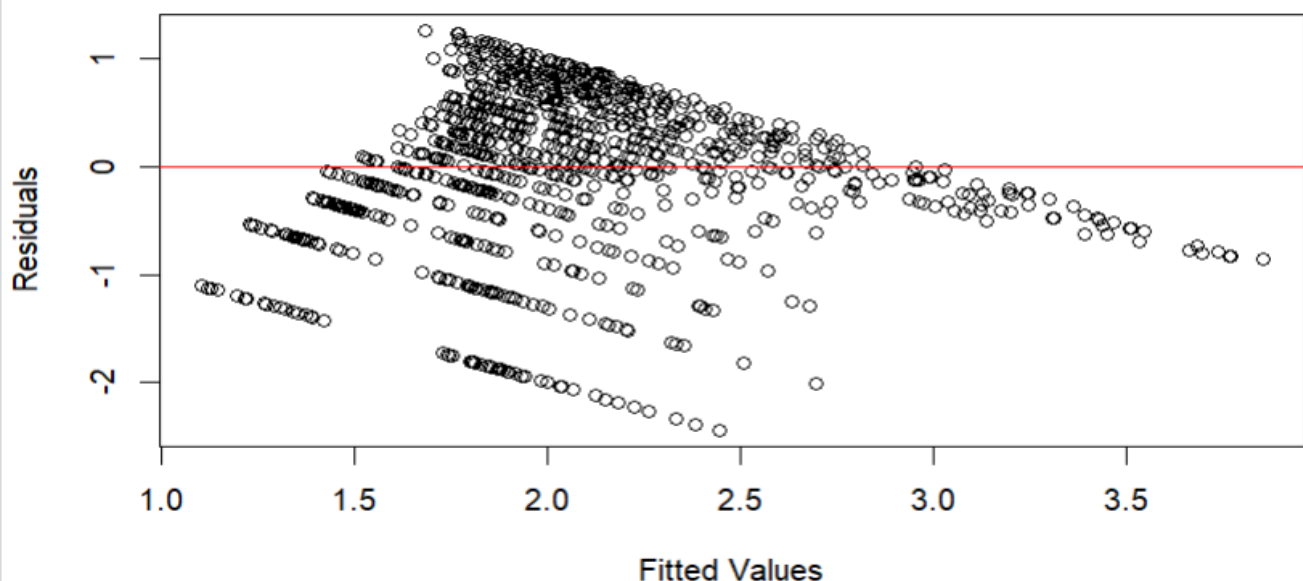


Figure 5: Residual diagnostics showing residuals centred around zero with no severe heteroscedasticity.

Minor patterns observed are consistent with discrete retail demand data and are acceptable in applied regression analysis.

8. Coefficient Visualization

Regression coefficients were visualized along with 95% confidence intervals using ggplot2.

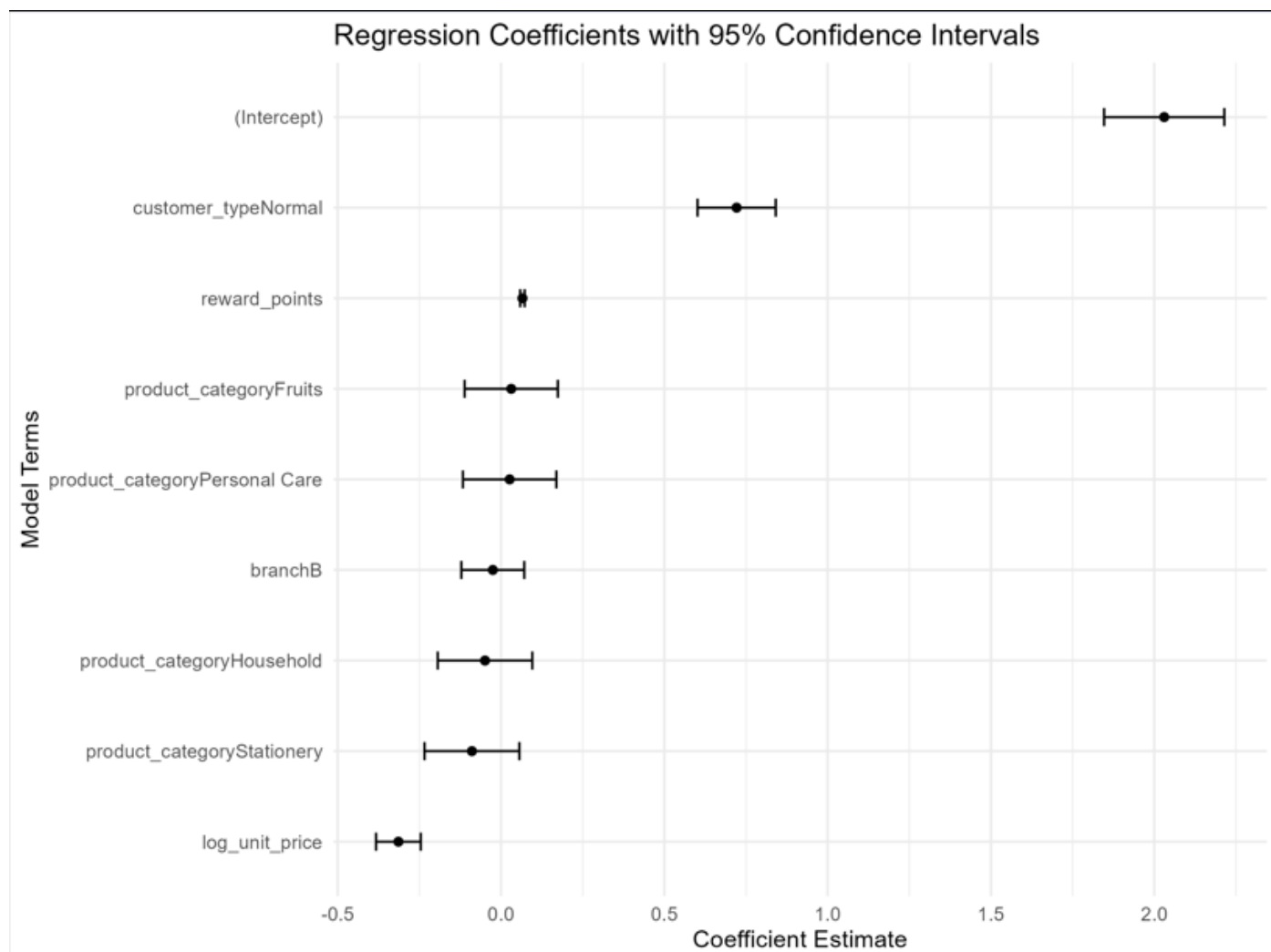


Figure 6: Regression coefficients with 95% confidence intervals highlighting the negative and statistically significant price elasticity.

This visualization clearly communicates both effect magnitude and estimation uncertainty.

9. Business Insights and Discussion

Key insights from the analysis include:

- Demand for supermarket products is price inelastic, allowing moderate price changes without substantial volume loss
- Loyalty mechanisms such as reward points significantly increase demand
- Demand responsiveness varies across product categories, suggesting segment-specific pricing strategies
- Branch-level effects are relatively small, indicating consistent pricing behaviour across locations

10. Conclusion

This project demonstrates the effective use of multivariate regression analysis to estimate price elasticity using real-world retail data. By applying a log-log model, validating assumptions, and visualizing coefficients, the study provides robust and interpretable insights into pricing behavior. The methodology aligns with industry practices and can be extended to promotional analysis, dynamic pricing, and demand forecasting.

11. Tools and Technologies Used

Tool / Package Purpose

R	Statistical analysis
tidyverse	Data manipulation
ggplot2	Data visualization
broom	Regression output formatting
car	Model diagnostics (VIF)

12. Deliverables Summary

- R Script File: price_elasticity_analysis.R
- Regression Summary Table: elasticity_regression_summary.csv
- Coefficient Plot: elasticity_coefficients_plot.png
- Interpretation Report: This document